arXiv:0704.0402v1 [math.AP] 3 Apr 2007

LOCATING THE PEAKS OF LEAST-ENERGY SOLUTIONS TO A QUASILINEAR ELLIPTIC NEUMANN PROBLEM

YI LI AND CHUNSHAN ZHAO

Abstract. In this paper we study the shape of least-energy solutions to the quasilinear problem $\varepsilon^m \Delta_m u - u^{m-1} + f(u) = 0$ with homogeneous Neumann boundary condition. We use an intrinsic variation method to show that as $\varepsilon \to 0^+$, the global maximum point P_ε of least-energy solutions goes to a point on the boundary $\partial\Omega$ at the rate of $o(\varepsilon)$ and this point on the boundary approaches to a point where the mean curvature of $\partial\Omega$ achieves its maximum. We also give a complete proof of exponential decay of least-energy solutions.

1. Introduction and statement of results

In this paper we study the shape of certain solutions to the following quasilinear elliptic Neumann problem:

$$\begin{cases} \varepsilon^m \Delta_m u - u^{m-1} + f(u) = 0, & u > 0 \text{ in } \Omega, \\ \dfrac{\partial u}{\partial \nu} = 0 & \text{on } \partial\Omega, \end{cases} \tag{1.1}$$

where m $(2 \le m < N)$ and $0 < \varepsilon \le 1$ are constants and $\Omega \subseteq \mathbb{R}^N$ $(N \ge 3)$ is a smooth bounded domain. The operator $\Delta_m u = \operatorname{div}(|\nabla u|^{m-2} \nabla u)$ is the m-Laplacian operator, and ν is the unit outer normal to $\partial\Omega$.

Problem (1.1) appears in the study of non-Newtonian fluids, chemotaxis and biological pattern formation. For example, in the study of non-Newtonian fluids, the quantity m is a characteristic of the medium: media with $m > 2$ are called dilatant fluids, and those with $m < 2$ are called pseudo-plastics. If $m = 2$, they are Newtonian fluids (see [3] and its bibliography for more backgrounds). For the case $m = 2$, (1.1) is also known as the stationary equation of the Keller–Segal system in chemotaxis [14] or the limiting stationary equation of the so-called Gierer–Meinhardt system in biological pattern formation (see [23]).

First let us recollect some results related to our problem. In a series of remarkable papers, C.-S. Lin, W.-M. Ni and I. Takagi [14], Ni and Takagi [17], [18] studied the Neumann problem for certain elliptic equations, including

$$\begin{cases} d\Delta u - u + u^p = 0, & u > 0 \text{ in } \Omega, \\ \dfrac{\partial u}{\partial \nu} = 0 & \text{on } \partial\Omega, \end{cases} \tag{1.2}$$

where $d > 0$, $p > 1$ are constants, and p is subcritical, i.e., $p < \frac{N+2}{N-2}$. First, Lin, Ni and Takagi [14] applied the mountain-pass lemma [1] to show the existence of

Key words and phrases. Quasilinear Neumann problem, m-Laplacian operator, least-energy solution, exponential decay, mean curvature.

a least-energy solution u_d to (1.2), by which is meant that u_d has the least energy among all solutions to (1.2) with the energy functional

$$I_d(u) = \int_\Omega \left(\frac{d}{2} |\nabla u|^2 + \frac{1}{2} u^2 - \frac{1}{p+1} u_+^{p+1} \right) dx$$

defined on $W^{1,2}(\Omega)$. Hereinafter $u_+ = \max\{u, 0\}$ and $u_- = \min\{u, 0\}$. Then in [17], [18], Ni and Takagi investigated the shape of the least-energy solution u_d as d becomes sufficiently small, and showed that u_d has exactly one peak (i.e., local maximum of u_d) at $P_d \in \partial\Omega$. Moreover, as d tends to zero, P_d approaches a point where the mean curvature of $\partial\Omega$ achieves its maximum. See [15] for a review in this field. Also see [16] for the critical case $p = \frac{N+2}{N-2}$, and [5], [6], [7], [8], [9] for existence and properties of multiple-peaks solutions to (1.2).

From now on we make some hypotheses on $f\colon \mathbb{R} \to \mathbb{R}$, as follows.

(1)

(H_2) $f(t) \equiv 0$ for $t \le 0$ and $f \in C^1(\mathbb{R})$.

(H_3) $f(t) = O(t^p)$ as $t \to \infty$ with $m - 1 < p < \dfrac{N(m-1)+m}{N-m}$.

(H_4) Let $F(t) = \displaystyle\int_0^t f(s)\, ds$. Then there exists a constant $\theta \in \left(0, \dfrac{1}{m}\right)$ such that $F(t) \le \theta t f(t)$ for $t > 0$.

(H_5) $\dfrac{f(t)}{t^{m-1}}$ is strictly increasing for $t > 0$ and $f(t) = O\left(t^{m-1+\delta}\right)$ as $t \to 0^+$ with a constant $\delta > 0$.

(H_6) Let $g(u) = \dfrac{(m-1)u^{m-1} - uf'(u)}{u^{m-1} - f(u)}$. Then $g(u)$ is non-increasing on (u_c, ∞), where u_c is the unique positive solution for $f(t) = t^{m-1}$.

Next we present some preliminary knowledge about least energy solutions of the following problem:

$$\begin{cases} \Delta_m u - u^{m-1} + f(u) = 0 & \text{in } \mathbb{R}^N \\ u > 0 & \text{in } \mathbb{R}^N \end{cases} \tag{1.3}$$

As before we define an "energy functional" $I{:}W^{1,m}(\mathbb{R}^N) \longrightarrow \mathbb{R}$ associated with (1.3) by

$$I(\tilde{v}) = \int_{\mathbb{R}^N} \left\{ \frac{1}{m} \left(\varepsilon^m |\nabla \tilde{v}|^m + |\tilde{v}|^m \right) - F(\tilde{v}_+) \right\} dx. \tag{1.4}$$

Next let us give a remark on ground states to the problem 1.3. Here by a ground state we mean a non-negative nontrivial C^1 distribution solution which tends to zero at ∞. For case $m = 2$, it is well known that the problem 1.3 has a unique ground state (up to translations) which is radially symmetric [4]. For case $2 < m < N$ uniqueness and radial symmetry of ground states are still open. But the Steiner symmetrization tells us the least-energy solutions must be radially symmetric (certainly least-energy solutions are ground states). Our assumptions guarantee that the uniqueness (up to translations) of radial ground states (see [20]), which implies the uniqueness of least-energy solutions of the problem (1.3). Exact exponential decay of radial ground states was given in [11], thus we have the following proposition about the unique radial least-energy solution to problem 1.3:

Proposition 1.1. *Under assumptions* (H_2)–(H_6)*, there is a unique least energy solution* $w(x)$ *for* (*1.3*) *satisfying:*

(*i*) *w is radial, i.e., $w(x) = w(|x|) = w(r)$ and $w \in C^1(\mathbb{R}^N)$ with $w(0) = \max_{X\in\mathbb{R}^N} w(x)$, $w'(0) = 0$ and $w'(r) < 0$, $\forall r > 0$.*

(*ii*) *$\lim_{r\longrightarrow\infty} w(r) r^{\frac{N-1}{m(m-1)}} e^{\left(\frac{1}{m-1}\right)^{\frac{1}{m}} r} = C_0 > 0$ for some constant C_0 and $\lim_{r\longrightarrow\infty} \frac{w'(r)}{w(r)} = -\left(\frac{1}{m-1}\right)^{\frac{1}{m}}$.*

Remark 1.1. A good example for $f(t)$ which satisfies all hypotheses (H_2)–(H_6) is $f(t) = t^p$ for $m - 1 < p < \dfrac{N(m-1)+m}{N-m}$.

Next we define an "energy functional" $J_\varepsilon \colon W^{1,m}(\Omega) \to \mathbb{R}$ associated with (1.1) by

$$J_\varepsilon(v) = \int_\Omega \left\{ \frac{1}{m}\left(\varepsilon^m |\nabla v|^m + |v|^m\right) - F(v_+) \right\} dx, \tag{1.5}$$

with $F(v_+) = \int_0^{v_+} f(s)\, ds$. Then the well-known mountain-pass lemma [1] implies that

$$c_\varepsilon = \inf_{h\in\Gamma} \max_{t\in[0,1]} J_\varepsilon(h(t)) \tag{1.6}$$

is a positive critical value of J_ε, where Γ is the set of all continuous paths joining the origin and a fixed nonzero element $e \in W^{1,m}(\Omega)$ such that $e \geq 0$ and $J_\varepsilon(e) \leq 0$. It turns out c_ε can also be characterized as follows:

$$c_\varepsilon = \inf_{u\in M_\varepsilon} J_\varepsilon(u)$$

with

$$M_\varepsilon = \left\{ u \in W^{1,m}(\Omega)\,;\, u \geq 0,\ u \not\equiv 0,\ \int_\Omega \left(\varepsilon^m |\nabla u|^m + u^m\right) dx = \int_\Omega f(u)\, u\, dx \right\}$$

or

$$c_\varepsilon = \inf\left\{ M[u] \mid u \in W^{1,m}(\Omega),\ u \not\equiv 0 \text{ and } u \geq 0 \text{ in } \Omega \right\} \tag{1.7}$$

with

$$M[u] = \sup_{t\geq 0} J_\varepsilon(tu).$$

Hence c_ε is the least positive critical value and a critical point u_ε of J_ε with critical value c_ε is called a least-energy solution. Notice also that if we let

$$c_* = \mathrm{I}(w) = \frac{1}{m}\int_{\mathbb{R}^N} \left(|\nabla w|^m + w^m\right) dx - \int_{\mathbb{R}^N} F(w)\, dx,$$

where w is the unique least energy solution of (1.3), then c_* can also be characterized as

$$c_* = \inf\left\{ M_*[v] \mid v \in W^{1,m}\left(\mathbb{R}^N\right),\ v \not\equiv 0 \text{ and } v \geq 0 \text{ in } \mathbb{R}^N \right\} \tag{1.8}$$

with

$$M_*[v] = \sup_{t\geq 0} I(tv).$$

We refer to Lemma 2.1 of [13] for the above characterizations.

Next we consider the following problem:

$v \in W^{1,m}\left(\mathbb{R}_+^N\right)$ with $\mathbb{R}_+^N = \left\{(x_1, \cdots, x_N) \in \mathbb{R}^N, x_N \geq 0\right\}$ and satisfies

$$\begin{cases} \Delta_m v - v^{m-1} + f(v) = 0, & v > 0 \text{ in } \mathbb{R}_+^N, \\ \dfrac{\partial v}{\partial x_N} = 0 & \text{on } x_N = 0. \end{cases} \tag{1.9}$$

The solutions of (1.9) can be characterized as critical points of the functional defined over $W^{1,m}\left(\mathbb{R}_+^N\right)$ as follows.

$$I_{\mathbb{R}_+^N}(\tilde{v}) = \frac{1}{m}\int_{\mathbb{R}_+^N} \left|\nabla \tilde{v}\right|^m + \tilde{v}^m)\, dx - \int_{\mathbb{R}_+^N} F(\tilde{v}_+)\, dx.$$

Similarly as above the least positive critical value C_* corresponding to least energy solutions of (1.9) can be characterized as

$$C_* = \inf_{\tilde{v} \in W^{1,m}\left(\mathbb{R}_+^N\right), \tilde{v} \geq 0, \tilde{v} \not\equiv 0} \sup_{t>0} I_{\mathbb{R}_+^N}(t\tilde{v}) \tag{1.10}$$

and moreover

$$C_* = \frac{1}{2} c_* \tag{1.11}$$

due to the boundary condition in (1.9) and the fact that w is radial and hence $\frac{\partial w}{\partial x_N} = 0$. We also refer to Lemma 2.1 of [13] for the above characterization of C_*. In Theorem 1.3 of [13], we proved the following theorem.

Theorem 1.1. *Under hypotheses* (H_2)–(H_6)*, let* u_ε *be a least-energy solution of* (1.1)*. Then all local maximum points(if more than one) of* u_ε *aggregate to a global maximum point* P_ε *at a rate of* $o(\varepsilon)$ *and* $dist(P_\varepsilon, \partial\Omega)/\varepsilon \to 0$ *as* $\varepsilon \to 0^+$*, where* $dist(\cdot,\cdot)$ *is the general distance function. Moreover, we have the following upper-bound estimate for* c_ε *as* $\varepsilon \to 0^+$:

$$c_\varepsilon \leq \varepsilon^N \left\{ \frac{1}{2} c_* - (N-1) \max_{P \in \partial\Omega} H(P)\, \gamma\varepsilon + o(\varepsilon) \right\}, \tag{1.12}$$

where $H(P)$ *denotes the mean curvature of* $\partial\Omega$ *at* P*,* $\gamma > 0$ *is a positive constant given by*

$$\gamma = \frac{1}{N+1} \int_{\mathbb{R}_+^N} \left|w'(|z|)\right|^m z_N\, dz. \tag{1.13}$$

Our goal in this paper is to locate the position on $\partial\Omega$ where the global maximum point P_ε of u_ε in $\overline{\Omega}$ approaches, provided ε is sufficiently small. For the case $m = 2$, Ni and Takagi [18] located the peak by linearizing the equation $d\Delta u - u + f(u) = 0$ around the ground state w. But this method fails for our problem with $m \neq 2$ due to the strong nonlinearity of the m-Laplacian operator $\Delta_m u = \mathrm{div}(|\nabla u|^{m-2} \nabla u)$. So we have to use the intrinsic variational method created by Del Pino and Felmer in [2] to attack it. We also give a complete proof of the exponential decay of the least-energy solution u_ε. We remark that our proof is complete and does not require the non-degeneracy of the unique radial least energy solution w as stated in Proposition 1.1, and hence it is different from Ni's and Takagi's work [17]. Now our results can be stated as follows:

Theorem 1.2. *Under hypotheses* (H_2)–(H_6)*, let* u_ε *be a least-energy solution of* (1.1) *and* $\tilde{P}_\varepsilon \in \partial\Omega$ *with* $dist(P_\varepsilon, \tilde{P}_\varepsilon) = dist(P_\varepsilon, \partial\Omega)$*. Then as* $\varepsilon \to 0^+$*, after passing to a sequence* $\tilde{P}_\varepsilon$ *approaches* $\bar{P} \in \partial\Omega$ *with*

(1)

(ii) $H\left(\bar{P}\right) = \max_{P\in\partial\Omega} H\left(P\right)$, *where* $H\left(P\right)$ *denotes the mean curvature of* $\partial\Omega$ *at* P *as stated before, and moreover*

(iii) *the associated critical value* c_ε *can be estimated as* $\varepsilon \to 0^+$ *as follows*:

$$c_\varepsilon = \varepsilon^N \left\{ \frac{1}{2} c_* - \left(N-1\right) H\left(\bar{P}\right) \gamma\varepsilon + o\left(\varepsilon\right) \right\}, \tag{1.14}$$

where c_*, γ *are as stated in Theorem* 1.1.

The organization of this paper is as follows: In Section 2, we will prove some lemmas which will be used in proving Theorem 1.2. The proof of Theorem 1.2 will be given in Section 3.

2. Some lemmas and exponential decay of u_ε

First we prove the following lemma related to exponential decay of the least-energy solution u_ε.

Lemma 2.1. *Let* ε *be sufficiently small and that the least-energy solution* u_ε *achieves its global maximum at some point* P_ε. *Then there exist two positive constants* c_3 *and* c_4 *independent of* u_ε *or* ε *such that*

$$\begin{aligned} u_\varepsilon\left(x\right) &\le c_3 \exp\left\{-c_4 \left|x - P_\varepsilon\right| / \varepsilon\right\} \\ \left|\nabla u_\varepsilon(x)\right| &\le c_3 \varepsilon^{-1} \exp\{-c_4 |x - P_\varepsilon| / \varepsilon\}. \end{aligned} \tag{2.1}$$

Before beginning to prove this lemma, we give a remark on it.

Remark 2.1. For the case $m = 2$, under the assumption of non-degeneracy of the linearized operator $\Delta - 1 + f'\left(w\right)$, where w is the unique ground state of (1.3), Ni and Takagi [18] showed that $u_\varepsilon\left(x\right)$ can be written as

$$u_\varepsilon\left(x\right) = w\left(x\right) + \varepsilon\phi_1\left(x\right) + o\left(\varepsilon\right) \tag{2.2}$$

and $\phi_1\left(x\right)$ enjoys the exponential-decay property ([18]). Clearly we cannot derive exponential decay of $u_\varepsilon\left(x\right)$ as stated in Lemma (2.1) from (2.2) even though both $w\left(x\right)$ and $\varepsilon\phi_1\left(x\right)$ have exponential decay property.

Proof of Lemma 2.1. Since $\partial\Omega$ is a smooth compact submanifold of R^N, it follows from the tubular neighborhood theorem [10] that there exists a constant $\omega\left(\Omega\right) > 0$ which depends only on Ω such that $\Omega_I = \left\{x \in \overline{\Omega}, d\left(x, \partial\Omega\right) < \omega\left(\Omega\right)\right\}$ is diffeomorphic to the inner normal bundle

$$\left(\partial\Omega\right)_I^N = \left\{(x, y) : x \in \partial\Omega, y \in \left(-\omega\left(\Omega\right), 0\right] \nu_x\right\},$$

here ν_x is the unit outer normal of $\partial\Omega$ at x, and the diffeomorphism is defined as follows: $\forall x \in \Omega_I$, there exists an unique $\hat{x} \in \partial\Omega$ such that $d\left(x, \hat{x}\right) = d\left(x, \partial\Omega\right)$, then $\Phi_* : x \longrightarrow \left(\hat{x}, -d\left(x, \hat{x}\right)\nu_{\hat{x}}\right)$. Moreover this diffeomorphism satisfies $\Phi_*|_{\partial\Omega} =$ Identity. Similarly, let $\Omega_O = \left\{x \in \mathbb{R}^N \setminus \Omega, d(x, \partial\Omega) < \omega\left(\Omega\right)\right\}$. Then Ω_O is diffeomorphic to the outer normal bundle

$$\left(\partial\Omega\right)_O^N = \left\{(x, y) : x \in \partial\Omega, y \in \left[0, \omega\left(\Omega\right)\right) \nu_x\right\},$$

and the diffeomorphism is given as follows. $\forall x \in \Omega_O$, there exists an unique $\bar{x} \in \partial\Omega$ such that $d(x, \bar{x}) = d\left(x, \partial\Omega\right)$, and then $\Phi_\# : x \longrightarrow \left(\bar{x}, d\left(x, \bar{x}\right)\nu_{\bar{x}}\right)$ and $\Phi_\#|_{\partial\Omega} =$ Identity. Note that $\left(\partial\Omega\right)_I^N$ is clearly diffeomorphic to $\left(\partial\Omega\right)_O^N$ via the following reflection $\Phi^* : \left(\partial\Omega\right)_I^N \longrightarrow \left(\partial\Omega\right)_O^N$ defined by $\Phi^*\left((x, y)\right) = (x, -y)$.

Therefore, $\Phi = \Phi_*^{-1} \circ \Phi^{*-1} \circ \Phi_\# : \Omega_O \longrightarrow \Omega_I$ is the desired diffeomorphism and $\Phi|_{\partial\Omega} =$ Identity. Moreover, if we let $x = \Phi(z) = (\Phi_1(z), \cdots, \Phi_N(z))$, $z \in \Omega_O$, and $z = \Psi(x) = \Phi^{-1}(x) = (\Psi_1(x), \cdots, \Psi_N(x))$, $x \in \Omega_I$, $g_{ij} = \sum_{k=1}^N \frac{\partial \Phi_k}{\partial z_i}\frac{\partial \Phi_k}{\partial z_j}$, $g^{ij} = \sum_{k=1}^N \frac{\partial \Psi_i}{\partial x_k}\frac{\partial \Psi_j}{\partial x_k}(\Phi(z))$, we have $g_{ij}|_{\partial\Omega} = g^{ij}|_{\partial\Omega} = \delta_{ij}$ with δ_{ij} being the Kronecker symbol. Denote $G = \left(g^{ij}\right)$ and $A = G - I$ with I being the $N \times N$ identity matrix, $g(x) = \det(g_{ij})$ and $\hat{u}_\varepsilon(x) = u_\varepsilon(\Phi(x))$ for $x \in \Omega_O$. Then $\hat{u}_\varepsilon(x)$ satisfies the following equations:

$$\begin{cases} \varepsilon^m \mathrm{L}\hat{u}_\varepsilon - \sqrt{g}\hat{u}_\varepsilon^{m-1} + \sqrt{g} f(\hat{u}_\varepsilon) \quad = 0, \quad \hat{u}_\varepsilon > 0 \quad \text{in} \quad \Omega_O \\ \dfrac{\partial \hat{u}_\varepsilon}{\partial \nu} = 0, \quad \text{on} \quad \partial\Omega, \end{cases}$$

where

$$\begin{aligned} \mathrm{L}\hat{u}_\varepsilon &= \sum_{i=1}^N \frac{\partial}{\partial x_i}\left\{\left[\sum_{s,l=1}^N g^{sl}\frac{\partial \hat{u}_\varepsilon}{\partial x_s}\frac{\partial \hat{u}_\varepsilon}{\partial x_l}\right]^{\frac{m-2}{2}} \sqrt{g}\sum_{j=1}^N g^{ij}\frac{\partial \hat{u}_\varepsilon}{\partial x_j}\right\} \\ &= \mathrm{Tr}\left\{D\left(\left[\nabla\hat{u}_\varepsilon G\left(\nabla\hat{u}_\varepsilon\right)^T\right]^{\frac{m-2}{2}} \sqrt{g}\left(\nabla\hat{u}_\varepsilon\right)G\right)\right\}, \end{aligned}$$

where Tr means taking the trace of a square matrix.

For $0 < \tilde{\gamma} \le \omega(\Omega)$, let $\Omega_{O\tilde{\gamma}} = \left\{x \in \overline{\Omega_O}, d(x, \partial\Omega) < \tilde{\gamma}\right\}$. We know $\|A\|_{C^0}$ can be made arbitrarily small by making $\tilde{\gamma}$ sufficiently small. Next we define

$$\bar{u}_\varepsilon = \begin{cases} u_\varepsilon(x), & x \in \Omega \\ \hat{u}_\varepsilon(x), & x \in \Omega_O, \end{cases}$$

$$\tilde{g}_{ij} = \begin{cases} \delta_{ij}, & x \in \overline{\Omega} \\ g_{ij}, & x \in \Omega_O, \end{cases}$$

$$\tilde{g}^{ij} = \begin{cases} \delta_{ij}, & x \in \overline{\Omega} \\ g^{ij}, & x \in \Omega_O, \end{cases}$$

and $\tilde{A}(x,\xi) = \left(\tilde{A}_1(x,\xi), \cdots, \tilde{A}_N(x,\xi)\right)$ for $\xi = (\xi_1, \cdots, \xi_N)$ with

$$\tilde{A}_i(x,\xi) = \left[\sum_{s,l=1}^N \tilde{g}^{sl}\xi_s\xi_l\right]^{\frac{m-2}{2}} \sqrt{\tilde{g}}\sum_{j=1}^N \tilde{g}^{ij}\xi_j$$

and $\tilde{g} = \det(\tilde{g}_{ij})$, $B(x,u) = \sqrt{\tilde{g}}\left(-u^{m-1} + f(u)\right)$. Then $\bar{u}_\varepsilon(x)$ satisfies

$$\varepsilon^m \operatorname{div}\left(\tilde{A}(x, \nabla\bar{u}_\varepsilon)\right) + B(x, \bar{u}_\varepsilon) = 0 \quad \text{in} \quad \overline{\Omega}\bigcup\Omega_O \tag{2.3}$$

in the weak sense.

For any ball $B_r(x_0) \subset \overline{\Omega} \bigcup \Omega_O$ with radius r and center $x_0 \in \Omega$, let $\rho = |x - x_0|$. Then for any smooth increasing function $\phi = \phi(\rho)$ we have

$$\begin{aligned}
&\left[(\nabla\phi)G(\nabla\phi)^T\right]^{\frac{m-2}{2}} \sqrt{g}(\nabla\phi)G \\
&= \left|\nabla\phi(I+A)(\nabla\phi)^T\right|^{\frac{m-2}{2}} \sqrt{\det(I+A)^{-1}}\nabla\phi(I+A) \\
&= |\nabla\phi|^{m-2}\nabla\phi + \int_0^1 \frac{d}{dt}\left(\left|\nabla\phi(I+tA)(\nabla\phi)^T\right|^{\frac{m-2}{2}} \sqrt{\det(I+tA)^{-1}}\nabla\phi(I+tA)\right)dt \\
&= |\nabla\phi|^{m-2}\nabla\phi \\
&+ \frac{m-2}{2}\int_0^1 \left|\nabla\phi(I+tA)(\nabla\phi)^T\right|^{\frac{m-4}{2}} \left((\nabla\phi)A(\nabla\phi)^T\right) \sqrt{\det(I+tA)^{-1}}\nabla\phi(I+tA)dt \\
&+ \int_0^1 \left|\nabla\phi(I+tA)(\nabla\phi)^T\right|^{\frac{m-2}{2}} \frac{\frac{d}{dt}\left(\det(I+tA)^{-1}\right)}{2\sqrt{\det(I+tA)^{-1}}}\nabla\phi(I+tA)dt \\
&+ \int_0^1 \left|\nabla\phi(I+tA)(\nabla\phi)^T\right|^{\frac{m-2}{2}} \sqrt{\det(I+tA)^{-1}}(\nabla\phi)Adt.
\end{aligned}$$

Therefore

$$\begin{aligned}
&\mathrm{Tr}\left[D\left(\left[(\nabla\phi)G(\nabla\phi)^T\right]^{\frac{m-2}{2}} \sqrt{g}(\nabla\phi)G\right)\right] \\
&\quad \le \frac{3}{2}\left|\left(\left|\phi'\right|^{m-2}\phi'\right)'\right| + \frac{3(N-1)}{2\rho}\left|\phi'\right|^{m-2}\phi' + K\left|\phi'\right|^{m-2}\phi'
\end{aligned} \tag{2.4}$$

by taking $\tilde{\gamma}$ sufficiently small, here $K > 0$ is a constant depending only on Ψ, hence only on Ω and $\phi' = \frac{d\phi(\rho)}{d\rho}$.

From now on $\tilde{\gamma} = \tilde{\gamma}(\Omega)$ is fixed such that (i) $\frac{3}{4} \le \sqrt{g} \le \frac{5}{4}$, (ii) (2.4) holds for any smooth increasing radial function $\phi(\rho)$ and (iii) $\frac{3}{4}|\xi|^m \le A(x,\xi)\cdot\xi \le \frac{5}{4}|\xi|^m$ for any $\xi = (\xi_1, \cdots, \xi_N)$. Denote $\Omega^{\tilde{\gamma}} = \Omega \cup \Omega_{O^{\tilde{\gamma}}}$.

Let $\Omega_\varepsilon = \frac{1}{\varepsilon}(\Omega - P_\varepsilon)$ and $u^\varepsilon(x) = u_\varepsilon(P_\varepsilon + \varepsilon x)$ for $x \in \Omega_\varepsilon$. Then u^ε is a solution to the following problem:

$$\begin{cases} \Delta_m u^\varepsilon - (u^\varepsilon)^{m-1} + f(u^\varepsilon) = 0, u^\varepsilon > 0 \quad \text{in} \quad \Omega_\varepsilon \\ \dfrac{\partial u^\varepsilon}{\partial n} = 0, \quad \text{on} \quad \partial\Omega_\varepsilon, \end{cases} \tag{2.5}$$

where n is the unit outer normal of $\partial\Omega_\varepsilon$. Similarly, let $\Omega_\varepsilon^{\tilde{\gamma}} = \frac{1}{\varepsilon}\left(\Omega^{\tilde{\gamma}} - P_\varepsilon\right)$ and $\bar{u}^\varepsilon(x) = \bar{u}_\varepsilon(P_\varepsilon + \varepsilon x)$ for $x \in \Omega_\varepsilon^{\tilde{\gamma}}$. Since $\bar{u}^\varepsilon$ converges to the unique radial least-energy solution w of (1.3) in $C^1_{loc}(\mathbb{R}^N) \cap W^{1,m}(\mathbb{R}^N)$ as $\varepsilon \to 0^+$ (see the proof of Theorem 1.2 of [13]) and w satisfies:

$$\text{(i)} \quad w \quad \text{is radial, i.e.,} w(x) = w(|x|) = w(r) > 0$$

$$\text{(ii)} \lim_{r\to\infty} w(r) r^{\frac{N-1}{m(m-1)}} e^{\left(\frac{1}{m-1}\right)^{\frac{1}{m}} r} = C_0 > 0$$

(see Theorem 1 of [11]) which yields $w(r) \le \kappa e^{-\mu r}$ for a constant $\kappa > 0$ and $\mu = \left(\frac{1}{m-1}\right)^{\frac{1}{m}}$. First we fix a constant $\eta > 0$ such that $\frac{1}{8}t^{m-1} > f(t)$ for $t \in (0, \eta]$. From hypothesis (H_5) it follows that such an η exists. Then there exist $\varepsilon_0 > 0$ sufficiently small and R_0 sufficiently large such that $4\kappa\exp\{-\mu R_0\} < \eta$ and

$\|\bar{u}^\varepsilon - w\|_{C^0(B_{R_0}(0)\cap\Omega_\varepsilon)} \le \kappa\exp\{-\mu R_0\}$, which yields

$$u^\varepsilon|_{(\partial B_{R_0}(0))\cap\Omega_\varepsilon} \le 2\kappa\exp\{-\mu R_0\}.$$

Note that

$$\begin{cases} \Delta_m u^\varepsilon - \frac{7}{8}(u^\varepsilon)^{m-1} = \frac{1}{8}(u^\varepsilon)^{m-1} - f(u^\varepsilon) > 0 \quad \text{in} \quad \Omega_\varepsilon \setminus B_{R_0}(0), \\ \frac{\partial u^\varepsilon}{\partial n} = 0 \quad \text{on} \quad \partial\Omega_\varepsilon \setminus B_{R_0}(0), \\ u^\varepsilon \le 2\kappa\exp\{-\mu R_0\} \quad \text{on} \quad \partial B_{R_0}(0)\cap\overline{\Omega_\varepsilon}. \end{cases}$$

Then we have

$$u^\varepsilon(x) \le 2\kappa\exp\{-\mu R_0\}, \quad \text{for} \quad x \in \Omega_\varepsilon \setminus B_{R_0}(0)$$

due to the strong maximum principle ([22]). We get by scaling back that

$$u_\varepsilon|_{\Omega\setminus B_{\varepsilon R_0}(0)} \le 2\kappa\exp\{-\mu R_0\}$$

and

$$\begin{aligned} u_\varepsilon(x) &\le w\left(\frac{|x|}{\varepsilon}\right) + \kappa\exp\{-\mu R_0\} \le \kappa\exp\{-\frac{\mu|x|}{\varepsilon}\} + \kappa\exp\{-\mu R_0\} \\ &\le 2\kappa\exp\{-\frac{\mu|x|}{\varepsilon}\} \end{aligned} \tag{2.6}$$

for $x \in \Omega \cap B_{\varepsilon R_0}(0)$.

From definition of $\bar{u}_\varepsilon$ we know

$$\bar{u}_\varepsilon(x) \le 2\kappa\exp\{-\frac{\mu\left(|x| - 2\mathrm{dist}\left(P_\varepsilon, \partial\Omega\right)\right)}{\varepsilon}\} \le 4\kappa\exp\{-\frac{\mu|x|}{\varepsilon}\} \quad \text{for} \quad x \in \Omega^{\tilde{\gamma}} \cap B_{\varepsilon R_0}(0)$$

for $\varepsilon \in (0, \varepsilon_0]$ with ε_0 sufficiently small due to the fact $\mathrm{dist}(P_\varepsilon, \partial\Omega) = o(\varepsilon)$ as $\varepsilon \to 0^+$. Note that

$$\sup_{\Omega^{\tilde{\gamma}}\setminus B_{\varepsilon R_0}(0)} \bar{u}_\varepsilon \le 4\kappa\exp\{-\mu R_0\}.$$

Choice of R_0 and $\tilde{\gamma}$ tells us for any $0 < t \le 4\kappa\exp\{-\mu R_0\}$

$$B(x,t) = \sqrt{\tilde{g}}\left(-t^{m-1} + f(t)\right) \le -\frac{1}{2}t^{m-1}.$$

$\forall x_0 \in \Omega \setminus B_{\varepsilon R_0}(0)$ and $B_r(x_0) \subset \Omega^{\tilde{\gamma}} \setminus B_{\varepsilon R_0}(0)$, define

$$\begin{aligned} \phi(x) = \phi(\rho) &= \phi(|x - x_0|) \\ &= \left\{\sup_{\Omega^{\tilde{\gamma}}\setminus B_{\varepsilon R_0}(0)} \bar{u}_\varepsilon\right\} \frac{\cosh\left(\frac{\lambda_*\rho}{\varepsilon}\right)}{\cosh\left(\frac{\lambda_* r}{\varepsilon}\right)}, \end{aligned}$$

where $\lambda_* > 0$ is a constant to be determined later. Simple calculations show that

(i) $\quad \phi'(\rho) > 0 \quad$ and $\quad \left(\left|\phi'\right|^{m-2}\phi'\right)' > 0;$

$$\begin{aligned} (ii) \quad & \varepsilon^m\left[\frac{3}{2}\left(\left|\phi'\right|^{m-2}\phi'\right)' + \frac{3(N-1)}{2\rho}\left|\phi'\right|^{m-2}\phi' + K\left|\phi'\right|^{m-2}\phi'\right] - \frac{1}{2}\phi^{m-1} \\ &= \left[\frac{3}{2}(m-1)(\lambda_*)^m\left(\tanh\left(\frac{\lambda_*\rho}{\varepsilon}\right)\right)^{m-2} + \frac{3}{2}(\lambda_*)^m\left(\frac{\tanh\left(\frac{\lambda_*\rho}{\varepsilon}\right)}{\left(\frac{\lambda_*\rho}{\varepsilon}\right)^{\frac{1}{m-1}}}\right)^{m-1}\right. \\ &\quad \left. + \varepsilon(\lambda_*)^{m-1}K\left(\tanh\left(\frac{\lambda_*\rho}{\varepsilon}\right)\right)^{m-1} - \frac{1}{2}\right]\phi^{m-1} \\ &\le 0 \end{aligned}$$

for any $0 < \lambda_* \leq \hat{\lambda}$, where $\hat{\lambda} > 0$ is a small constant depending only on m and Ω through K. We remark that we have used the fact $\max_{r\in[0,\infty)} \frac{\tanh r}{r^{\frac{1}{m-1}}} < \infty$ for $m \geq 2$. From now on we choose $\lambda_* = \hat{\lambda}$.

Therefore we have

$$\varepsilon^m \operatorname{div}(\tilde{A}(x, \nabla \bar{u}_\varepsilon)) - \frac{1}{2}\bar{u}_\varepsilon^{m-1} \geq 0 \quad in \quad B_r(x_0),$$
$$\varepsilon^m \operatorname{div}(\tilde{A}(x, \nabla \phi)) - \frac{1}{2}\phi^{m-1} \leq 0 \quad in \quad B_r(x_0).$$

Clearly

$$\phi|_{\partial B_r(x_0)} \geq \bar{u}_\varepsilon|_{\partial B_r(x_0)}.$$

Then from the Comparison Theorem (Theorem 10.1 of [19]) it follows that

$$\phi(x) \geq \bar{u}_\varepsilon(x) \quad in \quad B_r(x_0).$$

In particular, $\phi(x_0) \geq \bar{u}_\varepsilon(x_0)$. Thus we get

$$u_\varepsilon(x_0) \leq \left(\sup_{\Omega^{\tilde{\gamma}} \setminus B_{\varepsilon R_0}(0)} \bar{u}_\varepsilon \right) \exp\{-\frac{\lambda_* r}{\varepsilon}\}.$$

Choosing $r = d\left(x_0, \partial\left(\Omega^{\tilde{\gamma}} \setminus B_{\varepsilon R_0}(0)\right)\right)$ we get

$$u_\varepsilon(x_0) \leq 4\kappa \exp\{-\mu R_0 - \frac{\lambda_* r}{\varepsilon}\} \leq 2\kappa \exp\{-\frac{\tilde{\lambda}(\varepsilon R_0 + r)}{\varepsilon}\}$$

with $\tilde{\lambda} = \min\{\mu, \lambda_*\}$. Note that x_0 belongs to one of the following two cases:

$$(i) \quad d\left(x_0, \partial\left(\Omega^{\tilde{\gamma}} \setminus B_{\varepsilon R_0}(0)\right)\right) = d\left(x_0, \partial B_{\varepsilon R_0}(0)\right),$$
$$(ii) \quad d\left(x_0, \partial\left(\Omega^{\tilde{\gamma}} \setminus B_{\varepsilon R_0}(0)\right)\right) = d\left(x_0, \partial\Omega^{\tilde{\gamma}}\right).$$

For case (i) we have $d(x_0, P_\varepsilon) \leq \varepsilon R_0 + r$ and therefore

$$u_\varepsilon(x_0) \leq 4\kappa \exp\{-\frac{\tilde{\lambda} d(x_0, P_\varepsilon)}{\varepsilon}\}. \tag{2.7}$$

For case (ii) we have $r \geq \tilde{\gamma}$ and thus

$$\begin{aligned} u_\varepsilon(x_0) &\leq 4\kappa \exp\{-\tilde{\lambda}\frac{\varepsilon R_0 + r}{\varepsilon}\} \leq 4\kappa \exp\{-\frac{\tilde{\lambda}\tilde{\gamma}}{\varepsilon}\} \\ &\leq 4\kappa \exp\{-\tilde{\lambda}\frac{\tilde{\gamma}}{\operatorname{diam}(\Omega)} \cdot \frac{d(x_0, P_\varepsilon)}{\varepsilon}\}, \end{aligned} \tag{2.8}$$

Combining (2.6), (2.7) and (2.8) together and letting $\tilde{c_3} = 4\kappa$, $\tilde{c_4} = \min\{\mu, \tilde{\lambda}, \frac{\tilde{\lambda}\tilde{\gamma}}{\operatorname{diam}(\Omega)}\}$ yields

$$u_\varepsilon(x) \leq \tilde{c_3} \exp\{-\frac{\tilde{c_4}|x - P_\varepsilon|}{\varepsilon}\}. \tag{2.9}$$

Next we show the estimate for $|\nabla u_\varepsilon|$ holds. First from (2.5) it follows that

$$\Delta_m u^\varepsilon = (u^\varepsilon)^{m-1} - f(u^\varepsilon), u^\varepsilon > 0 \quad \text{in} \quad \Omega_\varepsilon \tag{2.10}$$

For $x \in \Omega_\varepsilon$ and $\operatorname{dist}(x, \partial\Omega_\varepsilon) \geq 1$, consider (2.10) in the unit ball centered at x, i.e., $B_1(x)$. Then by an $C^{1,\alpha}$ estimate (see [21], for example) there exists two constants

$C > 0$ and $\alpha^* \in (0,1)$ which are independent of ε such that

$$\begin{aligned} \|u^\varepsilon\|_{C^{1,\alpha^*}(B_{\frac{1}{2}}(x))} &\le C\left(\|u^\varepsilon\|_{L^\infty(B_1(x))} + \|\,(u^\varepsilon)^{m-1} - f(u^\varepsilon)\|^{\frac{1}{m-1}}_{L^\infty(B_1(x))}\right) \\ &\le c_3^* \exp\{-c_4^*|x-P_\varepsilon|\}, \end{aligned} \tag{2.11}$$

where we have used (2.9) and the fact that $u^\varepsilon(x) = u_\varepsilon(P_\varepsilon + \varepsilon x)$ for $x \in \Omega_\varepsilon$. Especially we have

$$|\nabla u^\varepsilon(x)| \le c_3^* \exp\{-c_4^*|x-P_\varepsilon|\}, \tag{2.12}$$

for $x \in \Omega_\varepsilon$ and $\mathrm{dist}(x, \partial\Omega_\varepsilon) \ge 1$. For $x \in \Omega_\varepsilon$ with $\mathrm{dist}(x,\partial\Omega_\varepsilon) < 1$. Let $x_0 \in \partial\Omega_\varepsilon$ be a point such that $\mathrm{dist}(x, x_0) = \mathrm{dist}(x, \partial\Omega_\varepsilon)$ and consider $\bar{u}^\varepsilon(x) = \bar{u}_\varepsilon(P_\varepsilon + \varepsilon x)$ in $B_2(x_0)$, the ball of radius 2 centered at x_0, then from (2.3) it follows that $\bar{u}^\varepsilon$ satisfies

$$\mathrm{div}\left(\tilde{A}(P_\varepsilon + \varepsilon x, \nabla \bar{u}^\varepsilon)\right) + B(P_\varepsilon + \varepsilon x, \bar{u}^\varepsilon) = 0 \quad \text{in} \quad B_2(x_0) \tag{2.13}$$

in the weak sense. Then applying an $C^{1,\alpha}$ estimate (see [21], for example) again yields as above that there exists two constants $C > 0$ and $\alpha_* \in (0,1)$ which are independent of ε such that

$$\begin{aligned} \|\hat{u}^\varepsilon\|_{C^{1,\alpha_*}(B_1(x_0))} &\le C\left(\|\hat{u}^\varepsilon\|_{L^\infty(B_2(x_0))} + \|B(P_\varepsilon + \varepsilon x, \hat{u}^\varepsilon)\|^{\frac{1}{m-1}}_{L^\infty(B_2(x_0))}\right) \\ &\le c_3^* \exp\{-c_4^*|x-P_\varepsilon|\} \end{aligned}$$

by adjusting c_3^* and c_4^* if it is necessary. Especially we have

$$|\nabla u^\varepsilon(x)| \le c_3^* \exp\{-c_4^*|x-P_\varepsilon|\}, \tag{2.14}$$

Thus combining (2.11) and (2.14) together and scaling back we have for $x \in \Omega$

$$|\nabla u_\varepsilon(x)| \le c_3^* \varepsilon^{-1} \exp\{-c_4^* \frac{|x-P_\varepsilon|}{\varepsilon}\}.$$

Proof of Lemma 2.1 is completed by letting $c_3 = \max\{\tilde{c}_3, c_3^*\}$ and $c_4 = \min\{\tilde{c}_4, c_4^*\}$. □

Remark 2.2. Our proof of the Lemma 2.1 with necessary minor modifications also works well for elliptic systems.

Next we present a lemma related to extensions of u_ε.

Lemma 2.2. *There exists a C^1-extension $\tilde{u}_\varepsilon$ of u_ε which has compact support in $\mathbb{R}^N$ and satisfies*

(1)

(ii) $\|\tilde{u}_\varepsilon\|_{W^{1,m}(\mathbb{R}^N)} \le c_5 \|u_\varepsilon\|_{W^{1,m}(\Omega)}$ *and* $\|\tilde{u}_\varepsilon\|_{C^1(\mathbb{R}^N)} \le c_5 \|u_\varepsilon\|_{C^1(\bar{\Omega})}$,

(iii) $\tilde{u}_\varepsilon$ *also has the exponential-decay property as stated in Lemma* 2.1*, i.e., there exists an absolute constant $\lambda \ge 1$ such that*

$$\begin{aligned} 0 \le \tilde{u}_\varepsilon &\le c_3 \lambda \exp\left\{-\frac{c_4}{\lambda}\frac{|x-P_\varepsilon|}{\varepsilon}\right\}, \\ |\nabla \tilde{u}_\varepsilon(x)| &\le c_3 \lambda \varepsilon^{-1} \exp\{-\frac{c_4}{\lambda}\frac{|x-P_\varepsilon|}{\varepsilon}\}. \end{aligned} \tag{2.15}$$

and

(1)

(iv) *there exists a positive constant* $\tilde{\delta} = \tilde{\delta}(\Omega)$ *such that for any* $P \in \partial\Omega$, $\tilde{u}_\varepsilon|_{B_{\tilde{\delta}}(P)\setminus\Omega}$ *is the reflection of* u_ε *through* $\partial\Omega$.

Proof. Let $\tilde{d} = d\left(\partial\Omega, \partial\Omega^{\tilde{\gamma}}\right)$ and $0 \le \varrho(x) \le 1$ be a smooth cut-off function such that $\varrho(x) \equiv 1$ for $x \in \{x \in \mathbb{R}^N, d(x,\Omega) \le \frac{\tilde{d}}{2}\}$ and $\varrho(x) \equiv 0$ for $x \in \mathbb{R}^N \setminus \left(\overline{\Omega} \bigcup \Omega_O\right)$. Then $\tilde{u}_\varepsilon = \varrho \bar{u}_\varepsilon$ satisfies (ii), (iii) and (iv) automatically. The proof of this lemma is completed. □

Similar to energy density introduced in [2], we define the energy density associated with (1.1) as follows:

$$E(w, y') = \left[\frac{1}{m}\left(|\nabla w|^m + w^m\right) - F(w)\right](y', 0) \quad \text{for } y' \in \mathbb{R}^{N-1}.$$

Then we have the following lemma.

Lemma 2.3. *Let* G *be a* C^2 *function in a neighborhood of the origin of* $\mathbb{R}^{N-1}$. *Then*

$$\sum_{i,j=1}^{N-1} \int_{\mathbb{R}^{N-1}} G_{ij}(0)\, y_i y_j E(w, y')\, dy' = 2\Delta G(0)\, \gamma,$$

where γ *is the constant defined in* (1.13), *and* $y' = (y_1, \ldots, y_{N-1})$, *and*

$$G_{ij}(0) = \frac{\partial^2 G}{\partial y_i \partial y_j}(0).$$

Proof. In Lemma 2.4 of [13], we showed that

$$\gamma = \frac{1}{2}\int_{\mathbb{R}^N_+} \left(\frac{1}{m}\left(|\nabla w|^m\right) + w^m - F(w)\right) z_N\, dz. \tag{2.16}$$

Next we introduce the polar coordinates

$$\begin{cases} z_1 = r\sin\theta_{N-1}\sin\theta_{N-2}\cdots\sin\theta_2\sin\theta_1, \\ z_2 = r\sin\theta_{N-1}\sin\theta_{N-2}\cdots\sin\theta_2\cos\theta_1, \\ z_3 = r\sin\theta_{N-1}\sin\theta_{N-2}\cdots\cos\theta_2, \\ \quad\vdots, \\ z_N = r\cos\theta_{N-1}, \end{cases}$$

and notice that

$$\mathbb{R}^N_+ = \Big\{ (r, \theta_1, \ldots, \theta_{N-1}) \mid r > 0,\ 0 \le \theta_1 < 2\pi, \\ 0 \le \theta_j < \pi \text{ for } j = 2, \ldots, N-2, \text{ and } 0 \le \theta_{N-1} < \frac{\pi}{2} \Big\}$$

and that

$$dz = r^{N-1}\sin\theta_2\sin^2\theta_3\cdots\sin^{N-2}\theta_{N-1}\, dr\, d\theta_1\cdots d\theta_{N-1}.$$

After elementary computations one obtains

$$\gamma = \frac{1}{2}\int_0^\infty \left(\frac{1}{m}\left(|w'(r)|^m + w^m(r)\right) - F(w(r))\right) r^N\, dr \cdot \omega_{N-2}, \tag{2.17}$$

where ω_{N-2} is the volume of the unit ball in $\mathbb{R}^{N-2}$. Here we used the fact that w is radially symmetric.

Using the radial symmetry of w again, we obtain

$$\begin{aligned}
(2.18)\qquad & \sum_{i,j=1}^{N-1}\int_{\mathbb{R}^{N-1}} G_{ij}(0)\, y_i y_j E(w,y')\, dy' \\
&= \sum_{i=1}^{N-1}\int_{\mathbb{R}^{N-1}} G_{ii}(0)\, y_i^2 E(w,y')\, dy' \\
&= \sum_{i=1}^{N} G_{ii}(0)\cdot\frac{1}{N-1}\int_{\mathbb{R}^{N-1}} |y'|^2 E(w,y')\, dy' \\
&= \Delta G(0)\cdot\int_0^\infty E(w,r)\, r^N\, dr\cdot\omega_{N-2},
\end{aligned}$$

where $E(w,r) = (1/m)\left(|w'(r)|^m + w^m(r)\right) - F(w(r))$. Comparing (2.17) and (2.18) yields

$$\sum_{i,j=1}^{N-1}\int_{\mathbb{R}^{N-1}} G_{ij}(0)\, y_i y_j E(w,y')\, dy' = 2\Delta G(0)\,\gamma.$$

The proof of Lemma 2.3 is completed. □

3. Proof of Theorem 1.2

With the help of the lemmas in Section 2, now we can give the proof of Theorem 1.2.

Proof of Theorem 1.2. Since as $\varepsilon\to 0^+$, $P_\varepsilon\to\partial\Omega$ at the rate of $o(\varepsilon)$, it follows that $d(P_\varepsilon,\tilde{P}_\varepsilon)/\varepsilon\to 0$, where $\tilde{P}_\varepsilon\in\partial\Omega$ is the closest point on $\partial\Omega$ to P_ε. then by passing to a sequence, $\tilde{P}_\varepsilon\to\bar{P}\in\partial\Omega$. After an ε-dependent rotation and translation, we may assume that $\tilde{P}_\varepsilon$ is at the origin and Ω can be described in a fixed cubic neighborhood V of $\bar{P}$ as the set

$$\{\,(x',x_N)\mid x_N>\psi_\varepsilon(x')\,\}\qquad \text{with } x'=(x_1,\ldots,x_{N-1}),$$

where ψ_ε is smooth, $\psi_\varepsilon(0)=0$, $\nabla\psi_\varepsilon(0)=0$. Furthermore, we may assume that ψ_ε converges locally in the C^2 sense to ψ, a corresponding parametrization at $\bar{P}$. Note that since $\tilde{P}_\varepsilon$ is the origin, so we have $P_\varepsilon/\varepsilon\to 0$ as $\varepsilon\to 0^+$. Thus we have $\tilde{u}^\varepsilon(x)=\tilde{u}_\varepsilon(\varepsilon x)=\tilde{u}_\varepsilon\left(\varepsilon\left(x-\frac{P_\varepsilon}{\varepsilon}\right)+P_\varepsilon\right)\to w(x)$ in $C^1_{loc}\left(\mathbb{R}^N\right)$ as $\varepsilon\to 0^+$. From the characterization of $c_\varepsilon=J_\varepsilon(u_\varepsilon)$ in Section 1, we have

$$\varepsilon^{-N}J_\varepsilon(u_\varepsilon)\ge\varepsilon^{-N}J_\varepsilon(tu_\varepsilon)=I_{\Omega_\varepsilon}(tu^\varepsilon)$$

for all $t>0$. Hereinafter

$$I_{\Omega_*}(v)=\frac{1}{m}\int_{\Omega_*}\left(|\nabla v|^m+|v|^m\right)dx-\int_{\Omega_*}F(v)\,dx.$$

Then

$$\begin{aligned}
(3.1)\quad I_{\Omega_\varepsilon}(tu^\varepsilon) &= I_{\Omega_\varepsilon}(t\tilde{u}^\varepsilon)\ge I_{\mathbb{R}^N_+\cap V_\varepsilon}(t\tilde{u}^\varepsilon)+I_{(\Omega_\varepsilon\cap V_\varepsilon)\setminus\mathbb{R}^N_+}(t\tilde{u}^\varepsilon)-I_{\left(\mathbb{R}^N_+\cap V_\varepsilon\right)\setminus\Omega_\varepsilon}(t\tilde{u}^\varepsilon)\\
&= \mathrm{I}+\mathrm{II}-\mathrm{III},
\end{aligned}$$

with $V_\varepsilon = \frac{1}{\varepsilon}V$. Let us choose $t = t_\varepsilon$ so that $I_{\mathbb{R}^N_+}(t\tilde{u}^\varepsilon)$ maximizes in t. Then from the definition of C_* in (1.10), equality (1.11) and Lemma 2.2 it follows that

$$\mathrm{I} = I_{\mathbb{R}^N_+ \cap V_\varepsilon}(t_\varepsilon \tilde{u}^\varepsilon) \geq \frac{c_*}{2} + O\left(e^{-c_6/\varepsilon}\right)$$

for some constant $c_6 > 0$ independent of ε. Next we give an estimate of t_ε.

Lemma 3.1. *There is a unique $t_\varepsilon \in (0,\infty)$ such that*

$$\frac{1}{m}\int_{\mathbb{R}^N_+} t_\varepsilon^m \left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) dx - \int_{\mathbb{R}^N_+} F\left(t_\varepsilon \tilde{u}^\varepsilon\right) dx$$
$$= \sup_{t\geq 0}\left[\frac{1}{m}\int_{\mathbb{R}^N_+} t^m \left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) dx - \int_{\mathbb{R}^N_+} F\left(t\tilde{u}^\varepsilon\right) dx\right],$$

and moreover

$$t_\varepsilon = 1 + o(1) \quad \text{as } \varepsilon \to 0^+. \tag{3.2}$$

Proof. Under assumption (H_5), the existence and uniqueness of t_ε can be proved similarly to the proof of Lemma 2.1 of [13]. Here we only need show (3.2). Let

$$h_\varepsilon(t) = \frac{t^m}{m}\int_{\mathbb{R}^N_+} \left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) dx - \int_{\mathbb{R}^N_+} F\left(t\tilde{u}^\varepsilon\right) dx. \tag{3.3}$$

Then

$$\begin{aligned} h'_\varepsilon(t) &= t^{m-1}\int_{\mathbb{R}^N_+} \left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) dx - \int_{\mathbb{R}^N_+} \tilde{u}^\varepsilon f\left(t\tilde{u}^\varepsilon\right) dx \\ &= t^{m-1}\int_{\mathbb{R}^N_+} \left(|\nabla w|^m + w^m\right) dx - \int_{\mathbb{R}^N_+} w f(tw)\, dx + o(1), \end{aligned} \tag{3.4}$$

here we have used the exponential decay of $\tilde{u}_\varepsilon$ in Lemma 2.2, exponential decay of w and $\tilde{u}^\varepsilon \to w$ in $C^1_{\mathrm{loc}}\left(\mathbb{R}^N\right)$ as $\varepsilon \to 0^+$. Moreover the term $o(1) \to 0$ uniformly in t on each compact interval as $\varepsilon \to 0^+$. (3.3) tells us $h_\varepsilon(1) = \frac{1}{2}c_* + o(1)$, which yields that t_ε is bounded and away from 0. Also from (3.4) it follows that

$$\begin{aligned} h'_\varepsilon(t) &= t^{m-1}\int_{\mathbb{R}^N_+} w f(w)\, dx - \int_{\mathbb{R}^N_+} w\, f(tw)\, dx + o(1) \\ &= t^{m-1}\int_{\mathbb{R}^N_+} w^m \left(\frac{f(w)}{w^{m-1}} - \frac{f(tw)}{(tw)^{m-1}}\right) dx + o(1). \end{aligned} \tag{3.5}$$

Therefore at $t = t_\varepsilon$ we have

$$\int_{\mathbb{R}^N_+} w^m \left(\frac{f(w)}{w^{m-1}} - \frac{f(t_\varepsilon w)}{(t_\varepsilon w)^{m-1}}\right) dx = o(1). \tag{3.6}$$

Since $f(t)/t^{m-1}$ is strictly increasing (see (H_5)) it follows from (3.6) that $t_\varepsilon = 1 + o(1)$. The proof of Lemma 3.1 is completed. □

Proof of Theorem 1.2 *continued.* Using again the exponential decay of u_ε in Lemma 2.1 and the expansion of t_ε in Lemma 3.1, we obtain

$$
\begin{aligned}
(3.7)\quad -\mathrm{II} &= -\int_{(\mathbb{R}^{N-1}\times\{0\})\cap V_\varepsilon} dy' \\
&\quad \cdot \int_{\frac{(\psi_\varepsilon(\varepsilon y'))_-}{\varepsilon}}^{0} \left[\frac{1}{m} t_\varepsilon^m \left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) - F\left(t_\varepsilon \tilde{u}^\varepsilon\right)\right](y', y_N)\, dy_N \\
&= -\left(1+o(1)\right) \int_{(\mathbb{R}^{N-1}\times\{0\})\cap(\Omega_\varepsilon\cap V_\varepsilon)} dy' \\
&\quad \cdot \int_{\frac{(\psi_\varepsilon(\varepsilon y'))_-}{\varepsilon}}^{0} \left[\frac{1}{m}\left(|\nabla u^\varepsilon|^m + (u^\varepsilon)^m\right) - F\left(u^\varepsilon\right)\right](y', y_N)\, dy_N.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
(3.8)\quad \mathrm{III} &= \left(1+o(1)\right) \int_{V_\varepsilon\cap(\mathbb{R}^{N-1}\times\{0\})} dy' \\
&\quad \cdot \int_{0}^{\frac{(\psi_\varepsilon(\varepsilon y'))_+}{\varepsilon}} \left[\frac{1}{m}\left(|\nabla \tilde{u}^\varepsilon|^m + (\tilde{u}^\varepsilon)^m\right) - F\left(\tilde{u}^\varepsilon\right)\right](y', y_N)\, dy_N.
\end{aligned}
$$

In above $a_+ = \max\{a, 0\}$, $a_- = \min\{a, 0\}$. Since $\psi_\varepsilon(0) = 0$, $\nabla\psi_\varepsilon(0) = 0$ and ψ_ε converges in the C^2 local sense to ψ, and $\tilde{u}^\varepsilon \to w$ in the C^1 local sense in $\mathbb{R}^N$ with uniform exponential decay with respect to ε, it follows from the dominated convergence theorem that

$$
\begin{aligned}
&\lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon}\left(-\mathrm{II} + \mathrm{III}\right) \\
&\quad = \frac{1}{2}\sum_{i,j=1}^{N=1} \int_{\mathbb{R}^{N-1}} \psi_{ij}(0)\, y_i y_j \left(\frac{1}{m}\left(|\nabla w|^m + w^m\right) - F(w)\right)(y', 0)\, dy' \\
&\quad = \Delta\psi(0)\,\gamma = (N-1)\, H\left(\bar{P}\right)\gamma \qquad \text{(by Lemma 2.3).}
\end{aligned}
$$

Thus we have

$$
c_\varepsilon \geq \varepsilon^N \left\{\frac{1}{2} c_* - (N-1)\, H\left(\bar{P}\right)\gamma\varepsilon + o(\varepsilon)\right\}.
$$

But (1.12) in Theorem 1.1 tells us

$$
c_\varepsilon \leq \varepsilon^N \left\{\frac{1}{2} c_* - (N-1) \max_{P\in\partial\Omega} H(P)\,\gamma\varepsilon + o(\varepsilon)\right\}.
$$

Therefore we get

(1)
(ii) $H\left(\bar{P}\right) = \max_{P\in\partial\Omega} H(P)$, which is (ii) of Theorem 1.2,

and

(1)
(iii) $c_\varepsilon = \varepsilon^N \left\{\frac{1}{2} c_* - (N-1)\, H\left(\bar{P}\right)\gamma\varepsilon + o(\varepsilon)\right\}$ as $\varepsilon \to 0^+$,

which is part (iii) of Theorem 1.2. The proof of Theorem 1.2 is completed. □

Acknowledgement. The authors want to give their thanks to anonymous referee for some helpful comments.

References

[1] A. Ambrosetti and P.H. Rabinowitz, *Dual variational methods in critical point theory and applications*, J. Functional Analysis **14** (1973), 349–381. MR 51 #6412

[2] M. Del Pino and P.L. Felmer, *Spike-layered solutions of singularly perturbed elliptic problems in a degenerate setting*, Indiana Univ. Math. J. **48** (1999), 883–898. MR 2001b:35027

[3] J.I. Díaz, *Nonlinear Partial Differential Equations and Free Boundaries, Vol. I: Elliptic Equations*, Research Notes in Mathematics, vol. 106, Pitman Advanced Publishing Program, Boston, 1985. MR 88d:35058

[4] B. Gidas, W.M. Ni, L. Nirenberg, Symmetry of positive solutions of nonlinear elliptic equations in $\mathbb{R}^N$, Advances in Math, Supplementary Studies **7A** (1981) 369-402. MR 84a:35083

[5] C. Gui, *Multipeak solutions for a semilinear Neumann problem*, Duke Math. J. **84** (1996), 739–769. MR 1997i:35052

[6] C. Gui and N. Ghoussoub, *Multi-peak solutions for a semilinear Neumann problem involving the critical Sobolev exponent*, Math. Z. **229** (1998), 443–474. MR 2000k:35097

[7] C. Gui and J. Wei, *Multiple interior peak solutions for some singularly perturbed Neumann problems*, J. Differential Equations **158** (1999), 1–27. MR 2000g:35035

[8] ———, *On multiple mixed interior and boundary peak solutions for some singularly perturbed Neumann problems*, Canad. J. Math. **52** (2000), 522–538. MR 2001b:35023

[9] C. Gui, J. Wei, and M. Winter, *Multiple boundary peak solutions for some singularly perturbed Neumann problems*, Ann. Inst. H. Poincaré Anal. Non Linéaire **17** (2000), 47–82. MR 2001a:35018

[10] M.-M. Hirsch, *Differential Topology*, Graduate Texts in Mathematics, vol. 33, Springer-Verlag, New York, 1976. MR 56# 6669

[11] Y. Li and C. Zhao, *A note on exponential decay properties of ground states for quasilinear elliptic equations*, Proc. Amer. Math. Soc. **133** (2005), 2005–2012. MR 2006a:35091

[12] ———, *On the structure of solutions to a class of quasilinear elliptic Neumann problems*, J. Differential Equations **212** (2005), 208–233. MR 2006b:35107

[13] ———, *On the shape of least-energy solutions for a class of quasilinear elliptic Neumann problems*, IMA Journal of Applied Mathematics 2007; doi: 10.1093/imamat/hx1032.

[14] C.-S. Lin, W.-M. Ni, and I. Takagi, *Large amplitude stationary solutions to a chemotaxis system*, J. Differential Equations **72** (1988), 1–27. MR 89e:35075

[15] W.-M. Ni,*Diffusion, cross-diffusion, and their spike-layer steady states*, Notices Amer. Math. Soc. 45 (1998), no. 1, 9–18. MR 99a:35132

[16] W.-M. Ni, X.B. Pan, and I. Takagi, *Singular behavior of least-energy solutions of a semilinear Neumann problem involving critical Sobolev exponents*, Duke Math. J. **67** (1992), 1–20. MR 93j:35081

[17] W.-M. Ni and I. Takagi, *On the shape of least-energy solutions to a semilinear Neumann problem*, Comm. Pure Appl. Math. **44** (1991), no. 7, 819–851. MR 92i:35052

[18] ———, *Locating the peaks of least-energy solutions to a semilinear Neumann problem*, Duke Math. J. **70** (1993), 247–281. MR 94h:35072

[19] P. Pucci and J. Serrin, *The strong maximum principle revisted*, J. Differential Equations **196** (2004), no. 1, 1-66. MR 2004k:35033

[20] J. Serrin and M.-X. Tang, *Uniqueness of ground states for quasilinear elliptic equations*, Indiana Univ. Math. J. **49** (2000), no. 3, 897–923. MR 2002d:35072

[21] P. Tolksdorf, *Regularity for a more general class of quasilinear elliptic equations*, J. Differential Equations **51** (1984), no. 1, 126–150. MR 85g:35047

[22] J. Vazquez, *A strong maximum principle for some quasilinear elliptic equations*, Appl. Math. Optim. **12** (1984), no. 3, 191–202. MR 86m:35018

[23] J. Wei, *On the boundary spike layer solutions to a singularly perturbed Neumann problem*, J. Differential Equations **134** (1997), 104–133. MR 98e:35076

Department of Mathematics, The University of Iowa, Iowa City, IA 52242

Department of Mathematics, Hunan Normal University, Changsha, Hunan
E-mail address: yi-li@uiowa.edu

Department of Mathematical Sciences, Georgia Southern University, Statesboro, GA 30460
E-mail address: czhao@GeorgiaSouthern.edu